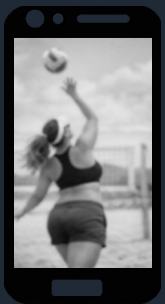
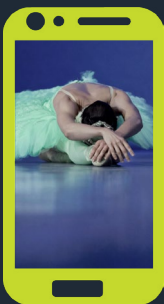
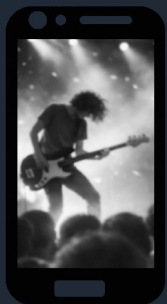
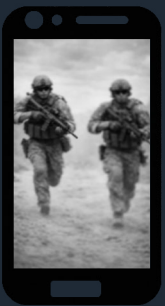
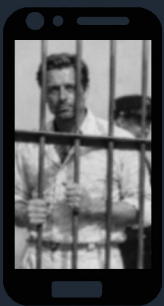
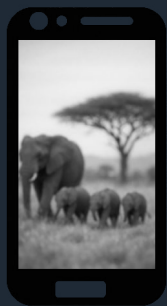
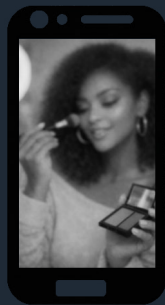
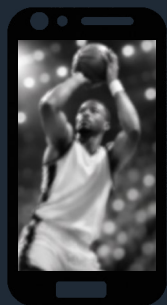


MOBILE PERSONALIZATION

The history and theory of
mobile streaming media



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JUSTIN GRANDINETTI
CHARLES ECENBARGER

MOBILE PERSONALIZATION

*The History and Theory of
Mobile Streaming Media*

by
Justin Grandinetti &
Charles Ecenbarger

Mobile Personalization: The History and Theory of Mobile Streaming Media

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INTRODUCTION

Mobile Matters

On April 6, 2020, an app launched in the Apple iOS and Android marketplaces in the United States, Canada, and select European countries with the goal of revolutionizing the streaming media experience by capturing mobile “on the go” audiences. Quibi (a fitting portmanteau of “quick bites”) was backed by a significant \$1.75 billion dollars in investment capital from Disney, 21st Century Fox, Sony, Viacom, Warner Media, Lionsgate, MGM, NBC, Alibaba, Wall Street banks, and the billionaire Walton family, and featured Hollywood-produced content with recognizable stars. The app appeared, at least on paper, to have what it takes to battle it out in the “streaming wars” against other established and upstart platforms. Setting Quibi apart was a mobile-only focus on which viewers could watch shows and movie-length content in ten to twenty-minute segments. Now just a few years later, Quibi is a defunct platform—its library of seventy-five programs sold for a comparatively meager \$100 million and rebranded as “Roku Originals.” Instead of taking a seat at the increasingly crowded table of mainstay streaming platforms, Quibi has the ignominy of being a mere footnote in the history of streaming (or perhaps a great trivia night question).

In the broadest sense, the expeditious rise and fall of Quibi is a reminder of the impermanence of technologies, apps, and platforms in a digital age. Few send faxes, for instance, in the era of email. Despite newfound hipster status, cassette tapes have become a rarity due to compact disks, which were subsequently displaced by MP3s, which were then later displaced by streaming. The erosion of Facebook’s (now Meta) social media dominance felt impossible until TikTok came around. In short, the technologies, apps, and platforms we rely on today can become obsolete or replaced tomorrow.

Returning to Quibi, many have attempted a postmortem on the platform—the incredibly poor timing of releasing an on-the-go streaming app coinciding with the immobility and lockdowns of a global pandemic, a lack of clear market, and limited mobile-only functionality are all plausible reasons for the platform’s failure. Executives at Quibi got a lot wrong when it came to how much their product would revolutionize the streaming world; however, the people behind Quibi *did* correctly identify the growing prevalence and importance of mobile streaming around the world.

This book isn’t about Quibi, but instead what Quibi was about—the ever-expanding practices of streaming media via mobile devices. Our aim is to untangle the complexity of the contemporary landscape of mobile streaming media. To do so, we consider both the past and present. Emphasis on current mobile streaming means considering the changing mobile devices, new generations of wireless infrastructures, and innovative platforms and applications that make these sociotechnical practices possible. We also want to assess how this moment of mobile streaming came about in the first place, which requires attention to both antecedent technologies and the trajectory of a cultural desire for a mobile and portable personalization and monetization of ever-growing times and spaces. We aim to define the characteristics of contemporary mobile streaming. By extension, we take seriously the challenge of telling a more complete story about mobile streaming by bringing together scholarship on streaming, mobility and mobile media, infrastructure, platforms, screens, and spatial relations. Put perhaps most simply, this book is about how and why we have come to engage in mobile streaming in a complex and ever-shifting sociotechnical landscape.

Just how ubiquitous are the wide variety of mobile streaming practices? While it’s difficult to quantify exactly how much streaming occurs through mobile devices around the world, surveys and studies have shown growth in mobile engagement year after year. Ownership of mobile devices has been overtaking fixed alternatives for some time. As of 2019, just over 51% of the 3.9 billion worldwide mobile users accessed the web solely through smartphones.¹ These numbers have continued to grow steadily each year, as Cisco estimates that over 70% of the global population (roughly 5.7 billion people) have mobile web connectivity as of 2023.² Due to this reliance on mobile media, some estimates, like that from a 2018 report by Intel,

¹ Lucy Handley, “Nearly Three Quarters of the World Will Use Just Their Smartphones to Access the Internet by 2025,” *Yahoo! Finance*, January 24, 2019.

² “Cisco Annual Internet Report (2018–2023),” Cisco, March 9, 2020.

predict that video will account for 90% of all 5G wireless traffic by 2028.³ It should come as little surprise, then, that official data from popular platforms conforms to these predictions. Roughly 90% of Twitter video views and 70% of YouTube content are viewed through mobile devices.⁴ The most downloaded app of 2020 was the short-form mobile streaming social media app TikTok, with over two billion global downloads and 732 million monthly active users, who engage with the app an average of nineteen times and eighty-nine minutes per day.⁵ Taken as a totality, it is clear that mobile matters now more than ever.

The fact that individuals are streaming more and more content from mobile devices is not surprising, considering the host of options available for use on smartphones and tablets. Platforms like Spotify, Tidal, and Apple Music allow users to stream algorithmically curated playlists and suggested podcasts to their smartphones. Audio-visual content providers like Netflix, Hulu, and Disney+ have increasingly optimized their movie and televisual content for mobile functionality. Video conferencing and video calls through apps such as FaceTime, Zoom, and Google Meetings, which saw a marked increase in use during the Covid-19 pandemic, have become integrated into the regular fabric of life. The previously mentioned TikTok has quickly threatened the grip of established social media giants like Facebook and Instagram—something that seemed near impossible only a few years ago. Feeling the pressure, Instagram has launched a clone of TikTok via Instagram Reels, whereas other platforms like Twitter and YouTube increasingly emphasize mobile engagement. So inescapable is mobile streaming that even the popular dating app Tinder has experimented with its own interactive, live streamed TV-like series. What these highly varied examples make clear is that it's hard to draw clear boundaries around what mobile streaming can be and that an emphasis on streaming through mobile devices has taken a prominent role in the contemporary media landscape.

³ "Intel Study Finds 5G Will Drive \$1.3 Trillion in New Revenues in Media and Entertainment Industry by 2028," *Intel Newsroom*, October 11, 2018.

⁴ "New Research: Twitter Users Love to Watch, Discover, and Engage with Video," *X Blog*, Twitter, March 13, 2015; "Marketing Strategies," Think with Google, Google, April–June 2018.

⁵ Rebecca Bellan, "The Top Social Media Apps of 2020, According to Apptopia," *Forbes*, December 3, 2020; Tim Ingham, "TikTok Users Watch the App for an Average of 89 Minutes per Day—and There's 732M of Them Worldwide," *Music Business Worldwide*, April 14, 2021; Tracy Qu, "TikTok and China Version Douyin Surpass Two Billion Download Milestone, Underlining Continued Appeal," *South China Morning Post*, April 30, 2020.

Of course, the growing adoption of mobile practices is well-documented by mobile media scholars attentive to the Global South. Mobile web connectivity is the normative means of access for most people around the world, largely because wireless infrastructures are easier and cheaper to create, while mobile devices are generally more affordable and useful than their fixed and immobile counterparts.⁶ Streaming platforms typically monetize through one of three strategies: the ad revenue-based advertising video-on-demand (AVOD), the monthly access membership of subscription video-on-demand (SVOD), or the individual purchase-based transactional video-on-demand (TVOD). Despite these differences in how platforms make money, these services are locked in an unending battle for viewers, their money, and their data. Countries in the Global South are no stranger to being used as a site of resource extraction for Western powers, and streaming media unfortunately has some parallels with colonial projects of the past, as platform providers have looked for new sources of revenue in these countries.⁷ For example, Netflix has trialed a mobile device-only subscription plan in select Asian countries, while the Walmart-owned and India-based app Flipkart launched in-app streaming in 2019 in response to tech giant Amazon's encroachment into Indian markets with discounted Prime subscriptions and streaming. Like the aforementioned experiments in mobile streaming, these examples are merely some of the many new ways that streaming has on a global scale become inexorably intertwined with mobile devices, technologies, infrastructures, apps, platforms, and new spatial and mobile practices.

This complexity also means that holistic scholarly accounts of mobile streaming are challenging. Instead, various media studies approaches provide a prism that refracts different elements of streaming. For instance, some scholars focus on streaming media industry shifts, including how streaming has displaced traditional cable television subscriptions, altered our orientation to owning music, and upended movie theater release cycles that have been in place for decades.⁸ Others consider the changing features of mobile devices that have enabled new bodily practices and relations to space.⁹ Mobile streaming also has overlap with studies of the different generations

⁶ Jonathan Donner, *After Access: Inclusion, Development, and a More Mobile Internet* (MIT Press, 2015), 25.

⁷ Paola Ricaurte, "Data Epistemologies, the Coloniality of Power, and Resistance," *Television & New Media* 20, no. 4 (2019): 351.

⁸ Amanda D. Lotz, *Media Disrupted: Surviving Pirates, Cannibals, and Streaming Wars* (MIT Press, 2021), 12.

⁹ Jason Farman, *Mobile Interface Theory: Embodied Space and Locative Media* (Taylor and Francis, 2021), 4.

of wireless infrastructure that facilitate all kinds of everyday streaming practices.¹⁰ Additionally, mobile streaming connects to academic conversations on platforms, including critical accounts of corporate consolidation, control, platformization, and datafication in a world where there's always "an app for that."¹¹ Complicating matters even further is how streaming and, by extension, mobile streaming fit within historical narratives. For example, scholars have considered the technological antecedents of television, film, and radio in relation to contemporary streaming.¹² Others still add to this archeological mix precursors that include the telephonoscope, long distance radio DXing, and even canals, aqueducts, and pipelines as models of infrastructures of flow.¹³ And, as if mobile streaming needed yet *another* dimension, these practices are facilitated by screen technologies, which have long been the subject of scholarship on the role of how screens structure spaces of everyday life.¹⁴ Taken together, mobile streaming brings together quite the big tent of interrelated scholarship.

Although we draw from this wealth of scholarship throughout the book, our lighthouse for this project is Raymond Williams's canonical work on the televisual.¹⁵ Williams is an enduring theorist of cultural studies, and his account of the emergence of television is often referenced for good reason. Perhaps what is most influential about Williams is that he rejects technological determinism—the notion that technology determines cultural change—to instead trace the emergence of television as a cultural technology. In doing so, Williams shows the formation of the televisual as intertwined with cultural needs and shaped by systems of power. Key here is the concept of mobile privatization, in which Williams theorizes how both the television and automobile were part of a deep contradiction of centralized transmission and privatized reception in the 1950s. During this time, families became

¹⁰ Heather A. Horst, "The Infrastructures of Mobile Media: Towards a Future Research Agenda," *Mobile Media & Communication* 1, no. 1 (2013): 147.

¹¹ Jean-Christophe Plantin et al., "Infrastructure Studies Meet Platform Studies in the Age of Google and Facebook," *New Media & Society* 20, no. 1 (2018): 298.

¹² Doron Galili, *Seeing by Electricity: Cinema, Moving Image Transmission, and the Emergence of Television, 1878–1939* (Duke University Press, 2020): 108.

¹³ Benjamin Burroughs, "A Cultural Lineage of Streaming," *Internet Histories* 3, no. 2 (2019): 2; Justin Grandinetti and Chris Ingraham, "Infrastructures of Flow: Streaming Media as Elemental Media," *Critical Studies in Media Communication* 39, no. 2 (2022): 2.

¹⁴ Jeremy Packer and Kathleen F. Oswald, "From Windscreen to Widescreen: Screening Technologies and Mobile Communication," *The Communication Review* 13, no. 4 (2010): 312; Lev Manovich, "A Screen's Genealogy," in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017), 125.

¹⁵ Raymond Williams, *Television: Technology and Cultural Form* (Fontana, 1974).

both more mobile and more private and isolated; the television emerged as a sociotechnical response to the need for contact outside the privatized space of the home.

We want to make clear that while streaming and mobile streaming are influenced by legacies of the televisual, streaming is far more than merely “the next stage of TV.” We do, nevertheless, take inspiration from Williams in a few crucial ways. First, Williams’s research on television eschews both determinism and “eureka” moments. Applied to our own project, mobile streaming did not just appear as the result of one singular thought, invention, or decision. This brings us to the second key inspiration from Williams’s work on the televisual: the emergence of technologies as intertwined with cultural and historical trajectories. Like Williams’s project, ours is a mix of historical and contemporary examples that help unpack how and why mobile streaming engagement has become so quotidian, embedded, and important.

Inspired by Williams’s mobile privatization, we call this book “mobile personalization.” With this term we refer to the historical trajectory of personalization via mobile and portable media to better understand how these technologies are critical components in transforming times and spaces in providing entertainment and information for users while simultaneously operating toward the goals of monetization and control. This is perhaps where we believe Williams’s influence to be the most important—such a focus permits us to consider historical antecedent technologies and discourses, as well as the interplay of technological, industry, and infrastructural changes that facilitate our contemporary mobile moment. In turn, we aim to decenter the primacy of one single technology (often the smartphone or tablet) to instead consider the emergence and development of our contemporary practices as shaped by a variety of discourses, imaginings, power structures, economic models, technological changes, and infrastructural development.

Such an approach helps consider mobile streaming not as a discrete present-day technology or event, but instead as a shifting sociotechnical constellation of humans, nonhumans, and technologies. In turn, this emphasis allows us to reflect on emergent relationships of power (contemporarily manifesting via the growing reach of consolidated tech companies and platform infrastructure providers), as well as the economic implications of mobile streaming. Stated differently, it took many different interdependent social and technical shifts to bring us the current mobile streaming landscape, and we aim to unpack how this moment came to be.

The following chapters employ a material-discursive framework for understanding the social-cultural-technical emergence of contemporary mobile streaming. We consider the discursive: the imaginings, discussions, and discourses on mobile and portable technologies so powerful in shaping what is seeable, sayable, and knowable about these multivariant practices over time. Furthermore, we are attentive to material changes that facilitate both historical antecedents and contemporary mobile streaming through the connected interplay of platforms, apps, infrastructures, devices, APIs, big data and artificial intelligence, codes, and users. Consequently, we are interested in mobile streaming not only in relation to mobile phones, tablets, and screens, but also the complex mix of historical trajectories, antecedent technologies, cultural imaginaries, and infrastructural technologies that enable these practices.

In chapter 1, we begin with the historical roots of mobile personalization. We first detail our approach to studying mobile streaming before providing an overview of technologies commonly considered precursors to streaming media, such as film and television. As our focus is on mobile streaming, we complicate these historical narratives with attention to the drive-in cinema and the portable television—technologies often overlooked or on the margins of commonly accepted historical narratives. We outline a brief history on the emergence of these technologies, with particular attention to the monetization of spaces and times through novel portable and mobile technologies.

Chapter 2 examines architectures of mobility. Here, we go beyond the smartphone to also consider the infrastructures and platforms that enable mobile streaming media. This includes the role of platform models and wireless infrastructures, and how platforms and infrastructures of mobile streaming have come to merge. Moreover, we consider some of the key technical underpinnings of mobile streaming like big data, algorithmic models, and APIs. Smartphones are certainly the point of access when it comes to streaming content, but infrastructures of data flows also play a pivotal role in creating, sending, and receiving streaming media.

Chapter 3 moves from the historical considerations of chapter 1 and architectural focus of chapter 2 to assess the varied practices of contemporary mobile streaming. From TikTok to Twitch, Instagram Reels to Apple Music, YouTube to videocalls, Switch to Stadia, we bring together the wide array of how mobile streaming has become embedded in everyday life around the world. We specifically focus on the different strategies of monetization emergent through mobile streaming. While once billed as a disruptor that

would end cable television, streaming has come to look (and cost) quite a bit like its predecessor. Additionally, the number of emergent streaming platforms belies the industry consolidation of big tech, both in the United States and around the world. We highlight numerous examples, including how US streaming platforms have attempted to grab international audiences as well as new geopolitical proxy battles between companies like Meta and TikTok.

Chapter 4 brings together our focus on the material-discursive historical trajectories of mobile streaming, as well as the host of architectures and monetization strategies to consider how mobile streaming necessitates a reimagining of streaming literature. Scholarship about streaming as spatially fixed has provided excellent conceptual anchors, yet it is critical to reassess and reconceptualize these attributes in the era of mobile streaming. Specifically, we revisit and refine some of the characteristics commonly associated with streaming media: mobile streaming flow, liveness and sharedness, screens and space, mobility and hybridity, and algorithms and big data.

We conclude our book with final thoughts on the past, present, and future of mobile streaming. Reflecting upon the interconnected strands of history, infrastructures, monetization, and platforms, we consider what's next for mobile streaming and streaming generally. Futurists failed to anticipate a landscape where we're glued to our phones more than ever before. We hope that by creating a more holistic look at the trajectories that brought us to this moment, we can, at the very least, better understand the present.